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Studierichting: Informatica  
Oefeningenreeks 1C nr. 58

$$\text{Opgave: } z^4 - z^3 + z^2 - z + 1 = 0$$

$$\Rightarrow (z + 1)(z^4 - z^3 + z^2 - z + 1) = 0$$

$$\Rightarrow z^5 - z^4 + z^3 - z^2 + z + z^4 - z^3 + z^2 - z + 1 = 0$$

$$\Rightarrow z^5 + 1 = 0 \Rightarrow z^5 = -1 + 0i$$

$$r = \sqrt{a^2 + b^2} = \sqrt{(-1)^2 + 0^2} = \sqrt{1} = 1$$

$$\tan \theta = \frac{b}{a} = 0 \Rightarrow \theta = \text{Bgtan}(0) = 0 + \pi, a < 0$$

$$w = \sqrt[5]{1} \operatorname{cis} \left( \frac{\pi + k \cdot 2 \cdot \pi}{5} \right), k \in \{0, 1, 2, 3, 4\}$$

$$w_0 = \operatorname{cis}(\pi) = -1$$

$$w_1 = \operatorname{cis} \left( \frac{3\pi}{5} \right) = \cos \left( \frac{3\pi}{5} \right) + i \cdot \sin \left( \frac{3\pi}{5} \right)$$

$$w_2 = \operatorname{cis} \left( \frac{5\pi}{5} \right) = -1$$

$$w_3 = \operatorname{cis} \left( \frac{7\pi}{5} \right) = \cos \left( \frac{7\pi}{5} \right) + i \cdot \sin \left( \frac{7\pi}{5} \right)$$

$$w_4 = \operatorname{cis} \left( \frac{9\pi}{5} \right) = \cos \left( \frac{9\pi}{5} \right) + i \cdot \sin \left( \frac{9\pi}{5} \right)$$

## References